

Interview-Style Questions Based on Conditional Statements

1. Check Even or Odd

Question: Determine whether a number is even or odd. **Explanation:** A number is even if it is divisible by 2. Otherwise, it's odd. - **Input:** Number = 6 - **Output:** Even number

```
n = 6
if n%2==0:
    print("Even")
else:
    print("Odd")
```

Output:

Even

2. Divisible by 5 but Not by 10

Question: Check if a number is divisible by 5 but not by 10. **Explanation:** Use modulo (%) to check if the number % 5 == 0 and number % 10 != 0. - **Input:** Number = 25 - **Output:** Satisfy

```
n=25
if n%5==0 and n%10!=0:
    print("Satisfy")
else:
    print("Not satisfy")
```

Output:

Satisfy

3. Biggest Among Two Numbers

Question: Find the biggest number among two. **Explanation:** Use comparison operators (>) to check which number is greater. - **Input:** A = 4, B = 7 - **Output:** Biggest is: 7

```
a=4
b=7
if a>b:
    print("biggest is:",a)
else:
    print("biggest is:",b)
```

Output:

biggest is: 7

4. Smallest Among Two Numbers

Question: Find the smallest number among two. **Explanation:** Use comparison operators (<) to find the smaller value. - **Input:** A = 4, B = 7 - **Output:** Smallest is: 4

```
A = 4
B = 7
if A<B:
    print("smallest is:",A)
else:
    print("smallest is:",B)
```

Output:

Smallest is: 4

5. Divisible by 2, 3, and 6

Question: Check if a number is divisible by 2, 3, and 6. **Explanation:** If a number is divisible by both 2 and 3, it is also divisible by 6. - **Input:** Number = 18 - **Output:** Satisfy

```
n=18
if n%2==0 and n%3==0 and n%6 ==0:
    print("Satisfy")
else:
    print("not Satisfy")
```

Output:

Satisfy

6. Voting Eligibility

Question: Check if a person is eligible to vote (age ≥ 18). **Explanation:** A person is eligible to vote if their age is 18 or above. - **Input:** Age = 19 - **Output:** Eligible to vote

```
age = 19
```

```
if age>=18:
```

```
    print("Eligible to vote")
```

```
else:
```

```
    print("not eligible to vote")
```

Output:

Eligible to vote

7. Student Pass/Fail Based on All Subjects ≥ 35

Question: Check if a student passed all subjects (maths, physics, chemistry). **Explanation:** Student passes only if marks in all subjects are 35 or more. - **Input:** Maths = 40, Physics = 36, Chemistry = 30 - **Output:** Fail

```
maths = 40
```

```
physics = 36
```

```
chemistry = 30
```

```
if maths >= 35 and physics >= 35 and chemistry >= 35:
```

```
    print("Pass")
```

```
else:
```

```
    print("Fail")
```

Output:

Fail

8. Student Pass if Passed Any One Subject (≥ 35)

Question: Check if the student passed at least one subject. **Explanation:** Use logical OR to check if any one subject has marks ≥ 35 . - **Input:** Maths = 20, Physics = 38, Chemistry = 25
- **Output:** Pass

```
maths = 20
physics = 38
chemistry = 25
if maths >= 35 or physics >= 35 or chemistry >= 35:
    print("Pass")
else:
    print("Fail")
```

Output:

Pass

9. Student Pass if Passed Any Two Subjects

Question: Check if the student passed any two out of three subjects. **Explanation:** Use a counter or logical conditions to verify two subjects ≥ 35 . - **Input:** Maths = 40, Physics = 20, Chemistry = 36 - **Output:** Pass

```
maths = 40
physics = 20
chemistry = 36
count = 0
if maths >= 35:
    count += 1
if physics >= 35:
    count += 1
if chemistry >= 35:
    count += 1
if count >= 2:
    print("Pass")
else:
```

```
print("Fail")
```

Output:

Pass

10. Biggest Among Three Numbers

Question: Find the biggest number among three. **Explanation:** Compare each pair of numbers using if-else conditions. - **Input:** A = 7, B = 4, C = 9 - **Output:** Biggest is: 9

```
a=7
```

```
b=4
```

```
c=9
```

```
if a>b and a>c:
```

```
    print("biggest is:",a)
```

```
elif b>a and b>c:
```

```
    print("biggest is:",b)
```

```
else:
```

```
    print("biggest is:",c)
```

Output:

Biggest is: 9

11. Smallest Among Three Numbers

Question: Find the smallest number among three. **Explanation:** Use comparison logic to determine the minimum value. - **Input:** A = 7, B = 4, C = 9 - **Output:** Smallest is: 4

```
a=7
```

```
b=4
```

```
c=9
```

```
if a<b and a<c:
```

```
    print("smallest is:",a)
```

```
elif b<a and b<c:
```

```
    print("smallest is:",b)
```

```
else:
```

```
print("smallest is:",c)
```

Output:

Smallest is: 4

12. Perfect Square or Not

Question: Check if a number is a perfect square. **Explanation:** A number is a perfect square if the square of its square root equals the number. - **Input:** Number = 49 - **Output:** Perfect square

```
num = 49
root = int(num ** 0.5)
if root * root == num:
    print("Perfect Square")
else:
    print("Not a Perfect Square")
```

Output:

Perfect Square

13. Cars Required for Members (Max 5 per car)

Question: Calculate how many cars are needed for a given number of people. **Explanation:** Divide total people by 5 and round up using ceiling logic. - **Input:** Members = 17 - **Output:** Cars needed = 4

```
members = 17
cars = members // 5
if members % 5 != 0:
    cars = cars + 1
print("Cars needed =", cars)
```

Output:

Cars needed = 4

14. Second Biggest Among Three Numbers

Question: Find the second largest number among three inputs. **Explanation:** Use sorting or nested conditions to find the second largest value. - **Input:** A = 10, B = 25, C = 18 - **Output:** Second biggest: 18

```
a = 10
b = 25
c = 18

if (a > b and a < c) or (a < b and a > c):
    print("Second biggest:", a)
elif (b > a and b < c) or (b < a and b > c):
    print("Second biggest:", b)
else:
    print("Second biggest:", c)
```

Output:

Second biggest: 18

15. Leap Year or Not

Question: Check if a given year is a leap year. **Explanation:** A year is a leap year if it is divisible by 4, and (not divisible by 100 unless divisible by 400). - **Input:** Year = 2024 - **Output:** Leap year

```
year = 2024

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap year")
else:
    print("Not Leap year")
```

Output:

Leap year